

Logical operators -

13 and -

Syntax: $\text{Condition 1} \text{ and } \text{Condition 2}$
 True $\Downarrow$ True

if both condition 1 $\xrightarrow{\text{True}}$ condition 2
are true then it returns true.

if-else block:

```
age = int(input("Enter your age"))  
nationality = input("Enter your  
nationality")
```

True/False

```
if (age >= 18) and (nationality ==  
"india");
```

```
print("you are eligible for voting")
```

else :

```
print("you are not eligible for voting")
```

and

Condition 1

True

True

False

False

Condition 2

True

False

True

False

→

→

→

→

True

False

False

False

2] or operator :

Syntax : Condition 1 or Condition 2

if any one condition is true,

it returns true

Or operator:-

Condition 1

Condition 2

True

True

True

True

False

True

False

True

True

False

False

False

```
weekend = True  
holiday = False
```

```
if weekend or holiday:  
    print("you can relax")
```

```
else:
```

```
    print("you need to work")
```

Conditional expression -

- This is the compact way of writing if-else statement.

- we can write if-else statement in one line

Syntax:

$\langle \text{exp1} \rangle$ if $\langle \text{conditional expression} \rangle$
else $\langle \text{exp2} \rangle$

Print("you are eligible for voting") if
age ≥ 18 else print("you
are not eligible for voting")

- age = 22 Cond. exp exp1

① if age >= 18 :

② print ("you are eligible
for voting")

③ else :

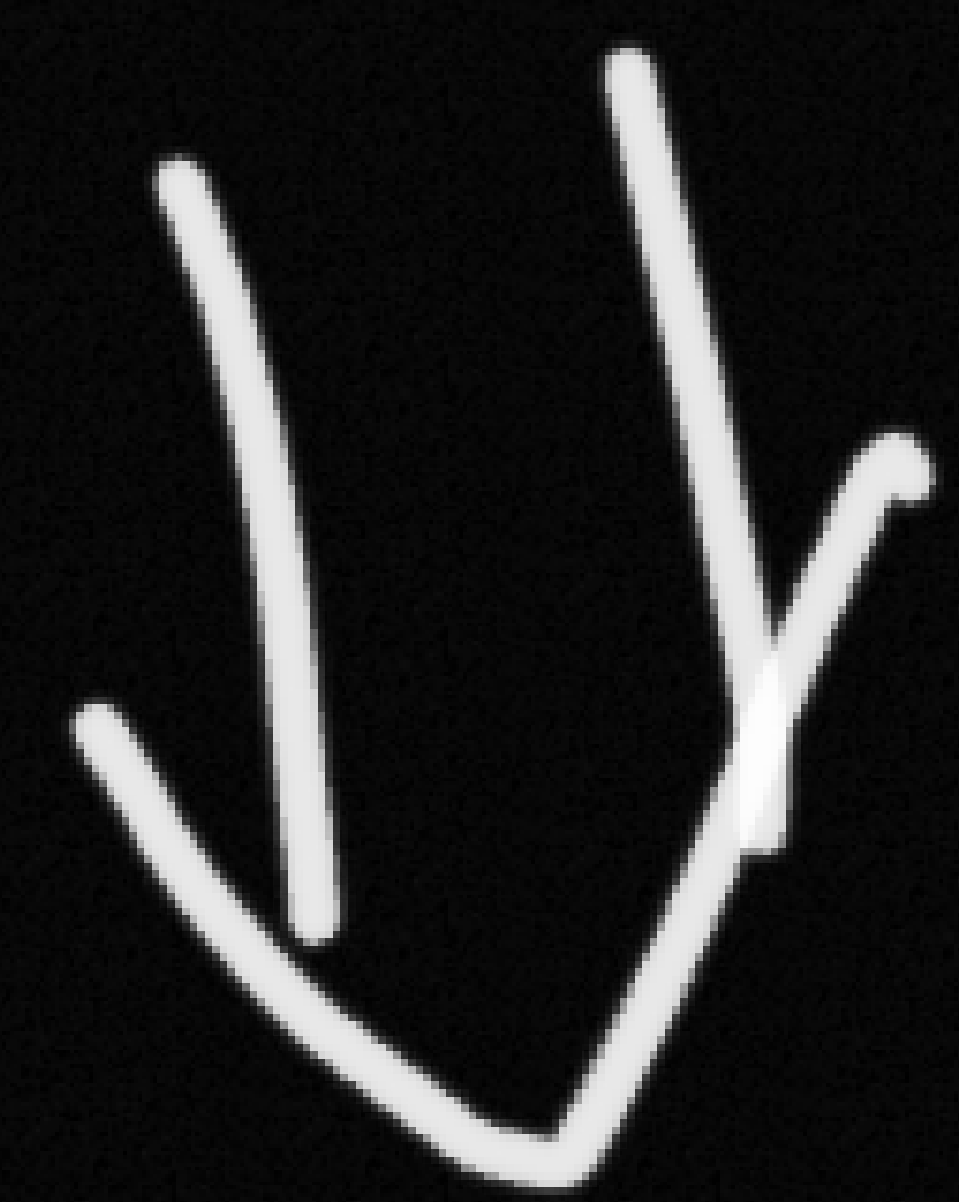
④ print ("you are not
eligible for voting")

exp2

Built in Functions:

- `all()` - similar to `and` operator
- `any()` - similar to `or` operator.

1] all() — all built-in function
returns true if all
conditions are true.



similar to and operator

age = 22

nationality = "indian"

result = all (age ≥ 18,
nationality == "indian")

print(result)

if result:
 print("you are eligible for voting")

else:
 print("you are not eligible for voting")

2] any() - any() built-in
function returns true if any
one condition is true.

weekend = True

holiday = False True False

result = any((weekend, holiday))

print(result)

if result:

 print("you can relax")

else:

 print("you need to work")

